

Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence

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Kate Crawford redefines artificial intelligence as neither truly "artificial" nor inherently "intelligent," but as a deeply material, embodied system shaped by natural resources, labor, and political structures. AI systems are extensions of societal power dynamics, optimized to serve dominant interests rather than operating as autonomous or rational entities.

AI is fundamentally political, reshaping how the world is perceived and understood. The book challenges readers to interrogate whose interests are being optimized and how power is concentrated in the hands of a few major companies, the "Great Houses of AI," which centralize computational and epistemological authority.

Crawford critiques the ecological and social costs of AI, noting the immense energy demands and carbon emissions of training large models. She links AI's environmental impact to broader industrial systems, such as shipping and mining, which sustain its operations, highlighting the hidden planetary costs of technological progress.

AI's integration into the workplace is often framed as collaborative, yet Crawford argues it forces workers into systems designed to quantify and control them. Power asymmetries inherent in these systems reduce genuine collaboration to compliance, privileging efficiency over fairness.

The book delves into the biases embedded in AI datasets, which are shaped by the socio-political contexts of their creators. Crawford underscores that data is not neutral but laden with worldviews, cultural assumptions, and systemic inequities, which are then amplified by AI systems.

Classification in AI, a core process of ordering the world into discrete categories, is critiqued for its reductive and often harmful consequences. Systems like ImageNet flatten human complexity into rigid taxonomies, perpetuating biases and reifying stereotypes under the guise of objectivity.

Crawford examines the problematic logic of data collection, where everything becomes raw material for computation. This extractive approach commodifies human experience and public knowledge, privileging profit-driven narratives while marginalizing diverse cultural and social realities.

The AI industry's practices of mass data extraction and surveillance reflect a broader privatization of knowledge. Public resources are transformed into proprietary systems that reinforce inequitable power structures, eroding public accountability and ethical standards.

The book also critiques the deployment of AI in militarized and governmental contexts. AI-driven surveillance and predictive policing reinforce systemic discrimination, creating feedback loops that perpetuate inequality while eroding civil liberties and public trust.

Crawford highlights the ethical void in AI research and development, where consent, caution, and accountability are often disregarded. The industry operates with minimal oversight, and ethical concerns are frequently subordinated to economic and political interests.

The narrative critiques the illusion of fairness in AI, particularly in emotion detection and hiring systems, which often rely on pseudoscientific assumptions. Crawford challenges the reduction of human emotions and identities to algorithmic predictions, arguing for a deeper consideration of cultural and contextual variability.

AI's expansive reach into education, healthcare, and governance reveals its entrenchment in capitalist and state power structures. Crawford warns against the unquestioned application of these technologies across diverse domains, emphasizing their profound societal implications.

The military-industrial nexus of AI is examined through projects like Project Maven, which align corporate and governmental interests in surveillance and warfare. These partnerships illustrate how AI systems are co-opted to serve geopolitical and economic agendas, often at the expense of human rights.

Crawford critiques the philosophical underpinnings of AI, which prioritize deterministic and reductionist approaches. She describes this as "enchanted determinism," where the mystical allure of AI's predictive power overshadows its real-world limitations and harms.

The narrative dismantles the myth of "AI ethics" frameworks, arguing that they often lack global representation and fail to address the systemic inequities AI perpetuates. True reform, Crawford argues, requires shifting the focus from ethics to power and centering marginalized communities.

The book calls for a reevaluation of AI's role in society, rejecting its portrayal as a neutral or universally beneficial tool. Crawford emphasizes that AI systems inherently reflect and amplify the inequalities and injustices of the societies in which they are developed.

Crawford critiques the epistemological violence of AI, where complex human realities are reduced to machine-readable proxies. She questions the legitimacy of AI's claims to objectivity and truth, arguing that its classifications often obscure more than they reveal.

The book argues that attempts to democratize AI are insufficient to dismantle the centralization of power it enables. Crawford posits that the tools of AI, designed for profit and control, cannot be easily repurposed for justice or equality without addressing their foundational biases.

Refusal and resistance are central to Crawford's vision of an alternative future for AI. She advocates rejecting the universal application of AI systems and focusing instead on grassroots justice movements that address labor, environmental, and racial inequities.

Crawford concludes by urging a critical reevaluation of the societal contracts underpinning AI. She calls for greater accountability and a shift toward technologies that prioritize human dignity, social equity, and ecological sustainability over profit and control.

Excerpts

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Stuart Russell and Peter Norvig state that AI is the attempt to understand and build intelligent entities. “Intelligence is concerned mainly with rational action,” they claim. “Ideally, an intelligent agent takes the best possible action in a situation.”¹⁸

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in this book I argue that AI is neither artificial nor intelligent. Rather, artificial intelligence is both embodied and material, made from natural resources, fuel, human labor, infrastructures, logistics, histories, and classifications. AI systems are not autonomous, rational, or able to discern anything without extensive, computationally intensive training with large datasets or predefined rules and rewards. In fact, artificial intelligence as we know it depends entirely on a much wider set of political and social structures. And due to the capital required to build AI at scale and the ways of seeing that it optimizes AI systems are ultimately designed to serve existing dominant interests. In this sense, artificial intelligence is a registry of power.

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At a fundamental level, AI is technical and social practices, institutions and infrastructures, politics and culture. Computational reason and embodied work are deeply interlinked: AI systems both reflect and produce social relations and understandings of the world.

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To understand how AI is fundamentally political, we need to go beyond neural nets and statistical pattern recognition to instead ask what is being optimized, and for whom, and who gets to decide. Then we can trace the implications of those choices.

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The field of AI is explicitly attempting to capture the planet in a computationally legible form. This is not a metaphor so much as the industry’s direct ambition. The AI industry is making and normalizing its own proprietary maps, as a centralized God’s-eye view of human movement, communication, and labor. Some AI scientists have stated their desire to capture the world and to supersede other forms of knowing. AI professor Fei-Fei Li describes her ImageNet project as aiming to “map out the entire world of objects.”²⁴ In their textbook, Russell and Norvig describe artificial intelligence as “relevant to any intellectual task; it is truly a universal field.”²⁵ One of the founders of artificial intelligence and early experimenter in facial recognition, Woody Bledsoe, put it most bluntly: “in the long run, AI is the only science.”²⁶ This is a desire not to create an atlas of the world but to be the atlas—the dominant way of seeing. This colonizing impulse centralizes power in the AI field: it determines how the world is measured and defined while simultaneously denying that this is an inherently political activity.

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In the case of AI, there is no singular black box to open, no secret to expose, but a multitude of interlaced systems of power. Complete transparency, then, is an impossible goal. Rather, we gain a better understanding of AI's role in the world by engaging with its material architectures, contextual environments, and prevailing politics and by tracing how they are connected.

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Simply put, artificial intelligence is now a player in the shaping of knowledge, communication, and power. These reconfigurations are occurring at the level of epistemology, principles of justice, social organization, political expression, culture, understandings of human bodies, subjectivities, and identities: what we are and what we can be. But we can go further. Artificial intelligence, in the process of remapping and intervening in the world, is politics by other means—although rarely acknowledged as such. These politics are driven by the Great Houses of AI, which consist of the half-dozen or so companies that dominate large-scale planetary computation.

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As Ursula Franklin writes, “The viability of technology, like democracy, depends in the end on the practice of justice and on the enforcement of limits to power.”

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The processing demands of training AI models, and thus their energy consumption, is still an emerging area of investigation. One of the early papers in this field came from AI researcher Emma Strubell and her team at the University of Massachusetts Amherst in 2019. With a focus on trying to understand the carbon footprint of natural language processing (NLP) models, they began to sketch out potential estimates by running AI models over hundreds of thousands of computational hours.⁵⁵ The initial numbers were striking. Strubell's team found that running only a single NLP model produced more than 660,000 pounds of carbon dioxide emissions, the equivalent of five gas-powered cars over their total lifetime (including their manufacturing) or 125 round-trip flights from New York to Beijing.

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OpenAI estimated that since 2012, the amount of compute used to train a single AI model has increased by a factor of ten every year.

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One container ship is estimated to emit as much pollution as fifty million cars,

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The common refrain for the expansion of AI systems and process automation is that we are living in a time of beneficial human-AI collaboration. But this collaboration is not fairly negotiated. The terms are based on a significant power asymmetry—is there ever a choice not to collaborate with algorithmic systems? When a company introduces a new AI platform, workers are rarely allowed to opt out. This is less of a collaboration than a forced engagement, where workers are expected to re-skill, keep up, and unquestioningly accept each new technical development.

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In her essay “How Silicon Valley Sets Time,” sociology professor Judy Wajcman argues that the aims of time-tracking tools and the demographic makeup of Silicon Valley are no coincidence.⁵³ Silicon Valley's elite workforce “is even younger, more masculine and more fully committed to working all hours,” while also creating productivity tools that are premised on a kind of ruthless, winner-takes-all race to maximal efficiency.⁵⁴ This means that young, mostly male engineers, often unencumbered by time-consuming familial or community responsibilities, are building the tools that will police very different workplaces, quantifying the productivity and desirability of employees. The workaholism and round-the-clock hours often glorified by tech start-ups become an implicit benchmark against which other workers are measured, producing a vision of a standard worker that is masculinized, narrow, and reliant on the unpaid or underpaid care work of others.

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I've looked at hundreds of datasets over years of research into how AI systems are built, but the NIST mug shot databases are particularly disturbing because they represent the model of what was to come. It's not just the overwhelming pathos of the images themselves. Nor is it solely the invasion of privacy they represent, since suspects and prisoners have no right to refuse being photographed. It's that the NIST databases foreshadow the emergence of a logic that has now thoroughly pervaded the tech sector: the unswerving belief that everything is data and is there for the taking. It doesn't matter where a photograph was taken or whether it reflects a moment of vulnerability or pain or if it represents a form of shaming the subject. It has become so normalized across the industry to take and use whatever

is available that few stop to question the underlying politics.

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The AI industry has fostered a kind of ruthless pragmatism, with minimal context, caution, or consent-driven data practices while promoting the idea that the mass harvesting of data is necessary and justified for creating systems of profitable computational “intelligence.” This has resulted in a profound metamorphosis, where all forms of image, text, sound, and video are just raw data for AI systems and the ends are thought to justify the means.

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it’s common practice for the first steps of creating a computer vision system to scrape thousands—or even millions—of images from the internet, create and order them into a series of classifications, and use this as a foundation for how the system will perceive observable reality. These vast collections are called training datasets, and they constitute what AI developers often refer to as “ground truth.”¹³ Truth, then, is less about a factual representation or an agreed-upon reality and more commonly about a jumble of images scraped from whatever various online sources were available.

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Training data, then, is the foundation on which contemporary machine learning systems are built.¹⁶ These datasets shape the epistemic boundaries governing how AI operates and, in that sense, create the limits of how AI can “see” the world. But training data is a brittle form of ground truth—and even the largest troves of data cannot escape the fundamental slippages that occur when an infinitely complex world is simplified and sliced into categories.

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The reliance on data over linguistic principles, however, presented a new set of challenges, for it meant that the statistical models were necessarily determined by the characteristics of training data. As a result, the size of the dataset became a central concern. . . . Larger datasets of observed outcomes not only improved the probability estimates for a random process, but also increased the chance that the data would capture more rarely-occurring outcomes. Training data size, in fact, was so central to IBM’s approach that in 1985, Robert Mercer explained the group’s outlook by simply proclaiming, “There’s no data like more data.”

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The Enron corpus has been cited in thousands of academic papers. Despite its popularity, it is rarely looked at closely: the New Yorker described it as “a canonic research text that no one has actually read.”²⁹ This construction of and reliance on training data anticipated a new way of doing things. It transformed the field of natural language processing and laid the foundations of what would become normal practice in machine learning.

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Text archives were seen as neutral collections of language, as though there was a general equivalence between the words in a technical manual and how people write to colleagues via email. All text was repurposable and swappable, so long as there was enough of it that it could train a language model to predict with high levels of success what word might follow another. Like images, text corpuses work on the assumption that all training data is interchangeable. But language isn’t an inert substance that works the same way regardless of where it is found. Sentences taken from Reddit will be different from those composed by executives at Enron. Skews, gaps, and biases in the collected text are built into the bigger system, and if a language model is based on the kinds of words that are clustered together, it matters where those words come from. There is no neutral ground for language, and all text collections are also accounts of time, place, culture, and politics. Further, languages that have less available data are not served by these approaches and so are often left behind.

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The approach of mass data extraction without consent and labeling by underpaid crowdworkers would become standard practice, and hundreds of new training datasets would follow ImageNet’s lead.

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From any seemingly innocuous and anonymized dataset can come many unexpected and highly personal forms of information, but this fact has not hampered the collection of images and text.

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Terms like “data mining” and phrases like “data is the new oil” were part of a rhetorical move that shifted the notion of data away from something personal, intimate, or subject to individual ownership and control toward something more inert and nonhuman. Data began to be described as a resource to be consumed, a flow to be controlled, or an investment to be harnessed.⁵⁶ The expression “data as oil” became commonplace, and although it suggested a picture of data as a crude material for extraction, it was rarely used to emphasize the costs of the oil and mining industries: indentured labor, geopolitical conflicts, depletion of resources, and consequences stretching beyond human timescales.

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If data is framed as oil, just waiting to be extracted, then machine learning has come to be seen as its necessary refinement process.

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As Fourcade and Healy observe, those who have the right data signals gain advantages like discounted insurance and higher standing across markets.⁵⁹ High achievers in the mainstream economy tend to do well in a data-scoring economy, too, while those who are poorest become targets of the most harmful forms of data surveillance and extraction.

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The great majority of university-based AI research is done without any ethical review process.

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Even though datasets in machine learning often come from and represent people and their lives, the research that used those datasets was seen more as a form of applied math with few consequences for human subjects. The infrastructures of ethics protections, like university-based institutional review boards (IRBs), had accepted this position for years.⁶³ This initially made sense; IRBs had been overwhelmingly focused on the methods common to biomedical and psychological experimentation in which interventions carry clear risks to individual subjects. Computer science was seen as far more abstract.

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A lyric from Tom Lehrer's satiric song about the wartime rocket scientist Wernher von Braun: "Once the rockets are up, who cares where they come down?" This separation of ethical questions away from the technical reflects a wider problem in the field, where the responsibility for harm is either not recognized or seen as beyond the scope of the research. As Anna Lauren Hoffman writes: "The problem here isn't only one of biased datasets or unfair algorithms and of unintended consequences. It's also indicative of a more persistent problem of researchers actively reproducing ideas that damage vulnerable communities and reinforce current injustices."

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The practices of data extraction and training dataset construction are premised on a commercialized capture of what was previously part of the commons. This particular form of erosion is a privatization by stealth, an extraction of knowledge value from public goods. A dataset may still be publicly available, but the metavalue of the data—the model created by it—is privately held.

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The new AI gold rush consists of enclosing different fields of human knowing, feeling, and action—every type of available data—all caught in an expansionist logic of never-ending collection. It has become a pillaging of public space.

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The way data is understood, captured, classified, and named is fundamentally an act of world-making and containment. It has enormous ramifications for the way artificial intelligence works in the world and which communities are most affected. The myth of data collection as a benevolent practice in computer science has obscured its operations of power, protecting those who profit most while avoiding responsibility for its consequences.

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Classification is an act of power, be it labeling images in AI training sets, tracking people with facial recognition, or pouring lead shot into skulls. But classifications can disappear, as Bowker and Star observe, "into infrastructure, into habit, into the taken for granted."¹⁶ We can easily forget that the classifications that are casually chosen to shape a technical system can play a dynamic role in shaping the social and material world.

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The vast majority of engineers hired by Amazon over ten years had been men, so the models they created, which were trained on the successful résumés of men, had learned to recommend men for future hiring. The employment practices of the past and present were shaping the hiring tools for the future. Amazon's system unexpectedly revealed the ways bias already existed, from the way masculinity is encoded in language, in résumés, and in the company itself. The tool was an intensification of the existing dynamics of Amazon and highlighted the lack of diversity across the AI industry past and present.²⁴

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The AI industry has traditionally understood the problem of bias as though it is a bug to be fixed rather than a feature of classification itself. The result has been a focus on adjusting technical systems to produce greater quantitative parity across disparate groups, which, as we'll see, has created its own problems.

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The foundational problems with IBM's approach to classifying diversity grow out of this kind of centralized production of identity, led by the machine learning techniques that were available to the team. Skin color detection is done because it can be, not because it says anything about race or produces a deeper cultural understanding. Similarly, the use of cranial measurement is done because it is a method that can be done with machine learning. The affordances of the tools become the horizon of truth. The capacity to deploy cranial measurements and digital epidermalization at scale drives a desire to find meaning in these approaches, even if this method has nothing to do with culture, heritage, or diversity. They are used to increase a problematic understanding of accuracy. Technical claims about accuracy and performance are commonly shot through with political choices about categories and norms but are rarely acknowledged as such.

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Machine learning systems can perform a type of induction, learning from specific examples (such as past résumés of job applicants) in order to decide which data points to look for in new examples (such as word groupings in résumés from new applicants). In such cases, the term "bias" refers to a type of error that can occur during this predictive process of generalization—namely, a systematic or consistently reproduced classification error that the system exhibits when presented with new examples. This type of bias is often contrasted with another type of generalization error, variance, which refers to an algorithm's sensitivity to differences in training data. A model with high bias and low variance may be underfitting the data—failing to capture all of its significant features or signals. Alternatively, a model with high variance and low bias may be overfitting the data—building a model too close to the training data so that it potentially captures "noise" in addition to the data's significant features.

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Every dataset used to train machine learning systems, whether in the context of supervised or unsupervised machine learning, whether seen to be technically biased or not, contains a worldview. To create a training set is to take an almost infinitely complex and varied world and fix it into taxonomies composed of discrete classifications of individual data points, a process that requires inherently political, cultural, and social choices. By paying attention to these classifications, we can glimpse the various forms of power that are built into the architectures of AI world-building.

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WordNet attempts to organize the entire English language into synonym sets, or synsets. The ImageNet researchers selected only nouns, with the idea that nouns are things that pictures can represent—and that would be sufficient to train machines to automatically recognize objects. So ImageNet's taxonomy is organized according to a nested hierarchy derived from WordNet, in which each synset represents a distinct concept, with synonyms grouped together (for example, "auto" and "car" are treated as belonging to the same set). The hierarchy moves from more general concepts to more specific ones. For example, the concept "chair" is found under artifact → furnishing → furniture → seat → chair. This classification system unsurprisingly evokes many prior taxonomical ranks, from the Linnaean system of biological classification to the ordering of books in libraries.

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Reducing humans into binary gender categories and rendering transgender people invisible or "deviant" are common features of classification schemes in machine learning. Os Keyes's study of automatic gender detection in facial recognition shows that almost 95 percent of papers in the field treat gender as binary, with the majority describing gender as immutable and physiological.⁴⁸ While some might respond that this can be easily remedied by creating more categories, this fails to address

the deeper harm of allocating people into gender or race categories without their input or consent.

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To impose order onto an undifferentiated mass, to ascribe phenomena to a category—that is, to name a thing—is in turn a means of reifying the existence of that category.

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To borrow an idea from linguist George Lakoff, the concept of an “apple” is a more nouny noun than the concept of “light,” which in turn is more nouny than a concept such as “health.”⁵¹ Nouns occupy various places on an axis from the concrete to the abstract, from the descriptive to the judgmental. These gradients have been erased in the logic of ImageNet. Everything is flattened out and pinned to a label, like taxidermy butterflies in a display case. While this approach has the aesthetics of objectivity, it is nonetheless a profoundly ideological exercise.

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ImageNet also contains categories that make no sense whatsoever for image classification such as Debtor, Boss, Acquaintance, Brother, and Color-Blind Person. These are all nonvisual concepts that describe a relationship, be it to other people, to a financial system, or to the visual field itself. The dataset reifies these categories and connects them to images, so that similar images can be “recognized” by future systems.

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Images—like all forms of data—are laden with all sorts of potential meanings, irresolvable questions, and contradictions. In trying to resolve these ambiguities, ImageNet’s labels compress and simplify complexity. The focus on making training sets “fairer” by deleting offensive terms fails to contend with the power dynamics of classification and precludes a more thorough assessment of the underlying logics.

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The idea that race and gender can be automatically detectable in machine learning is treated as an assumed fact and rarely questioned by the technical disciplines, despite the profound political problems this presents.

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Above all, these systems of classification have caused enormous harm to people, and the concept of a pure “race” signifier has always been in dispute. In her writing about race, Donna Haraway observes, “In these taxonomies, which are, after all, little machines for clarifying and separating categories, the entity that always eluded the classifier was simple: race itself. The pure Type, which animated dreams, sciences, and terrors, kept slipping through, and endlessly multiplying, all the typological taxonomies.”

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Machine learning systems are, in a very real way, constructing race and gender: they are defining the world within the terms they have set, and this has long-lasting ramifications for the people who are classified.

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At multiple levels, the act of defining categories and ideas of normalcy creates an outside: forms of abnormality, difference, and otherness. Technical systems are making political and normative interventions when they give names to something as dynamic and relational as personal identity, and they commonly do so using a reductive set of possibilities of what it is to be human. That restricts the range of how people are understood and can represent themselves, and it narrows the horizon of recognizable identities.

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Making these choices about which information feeds AI systems to produce new classifications is a powerful moment of decision making: but who gets to choose and on what basis? The problem for computer science is that justice in AI systems will never be something that can be coded or computed. It requires a shift to assessing systems beyond optimization metrics and statistical parity and an understanding of where the frameworks of mathematics and engineering are causing the problems.

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We are seeing strange, unpredictable twists as machine categories and people interact and change each other, as they try to find legibility in the shifting terrain, to fit the right categories and be spiked into the most lucrative feeds.

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The training sets that are made public—such as ImageNet, UTKFace, and DiF—give us some insight into the kinds of categorizations that are propagating across industrial AI systems and research practices. But the truly massive engines of classification are the ones being operated at a global scale by private technology companies, including Facebook, Google, TikTok, and Baidu. These companies operate with little oversight into how they categorize and target users, and they fail to offer meaningful avenues for public contestation. When the matching processes of AI are truly hidden and people are kept unaware of why or how they receive forms of advantage or disadvantage, a collective political response is needed—even as it becomes more difficult.

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Today affect recognition tools can be found in national security systems and at airports, in education and hiring start-ups, from systems that purport to detect psychiatric illness to policing programs that claim to predict violence.

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a comprehensive review of the available scientific literature on inferring emotions from facial movements published in 2019 was definitive: there is no reliable evidence that you can accurately predict someone's emotional state from their face.⁶

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Automated affect detection systems are now widely deployed, particularly in hiring. A startup in London called Human uses emotion recognition to analyze video interviews of job candidates. According to a report in the Financial Times, "The company claims it can spot the emotional expressions of prospective candidates and match them with personality traits"; the company then scores subjects on such personality traits as honesty or passion for a job.⁷ The AI hiring company HireVue, which lists among its clients Goldman Sachs, Intel and Unilever, uses machine learning to assess facial cues to infer people's suitability for a job. In 2014, the company launched its AI system to extract microexpressions, tone of voice, and other variables from video job interviews, which they used to compare job applicants against the company's top performers.⁸ In January 2016, Apple acquired the startup Emotient, which claimed to have produced software capable of detecting emotions from images of faces.⁹ Emotient grew out of academic research conducted at the University of California San Diego and is one of a number of startups working in this area.¹⁰ Perhaps the largest of these is Affectiva, a company based in Boston that emerged from academic work done at Massachusetts Institute of Technology. At MIT, Rosalind Picard and her colleagues were part of an emergent wider field known as affective computing, which describes computing that "relates to, arises from, or deliberately influences emotion or other affective phenomena."¹¹ Affectiva codes a variety of emotion-related applications, primarily using deep learning techniques. These range from detecting distracted and "risky" drivers on roads to measuring the emotional responses of consumers to advertising. The company has built what they call the world's largest emotion database, made up of over ten million people's expressions from eighty-seven countries.¹² Their monumental collection of videos of people emoting was hand labeled by crowdworkers based primarily in Cairo.¹³ Many more companies have now licensed Affectiva's products to develop everything from applications that assess job candidates to analyzing whether students are engaged in class, all by capturing and analyzing their facial expressions and body language.¹⁴ Beyond the start-up sector, AI giants like Amazon, Microsoft, and IBM have all designed systems for affect and emotion detection. Microsoft offers emotion detection in its Face API, which claims to detect what an individual is feeling across the emotions of "anger, contempt, disgust, fear, happiness, neutral, sadness, and surprise" and asserts that "these emotions are understood to be cross-culturally and universally communicated with particular facial expressions."¹⁵ Amazon's Rekognition tool similarly claims that it can identify "all seven emotions" and "measure how these things change over time, such as constructing a timeline of the emotions of an actor."

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Tomkins acknowledged that the interpretation of affective displays depends on individual, social, and cultural factors. He admitted that there were very different "dialects" of facial language in different societies. Even the forefather of affect research raised the possibility that recognizing affect and emotion depends on social and cultural context. The potential conflict between cultural dialects and a biologically based, universal language had enormous implications for the study of facial expression and later forms of emotion recognition. Given that facial expressions are culturally variable, using them to train machine learning systems would inevitably mix together all sorts of different contexts, signals, and expectations.

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When looking at these training sets, it is difficult to not be struck by how extreme they are: Incredible surprise! Abundant joy! Paralyzing fear! These subjects are literally making machine-readable emotion.

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Margaret Mead took a middle ground, emphasizing that there was no inherent contradiction between “the possibility that human beings may share a core of innate behaviors . . . and the idea that emotional expressions could, at the same time, be highly-conditioned by cultural factors.”

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Many datasets underlying the field are based on actors simulating emotional states, performing for the camera. That means that AI systems are trained to recognize faked expressions of feeling. Although AI systems claim to have access to ground truth about natural interior states, they are trained on material that is inescapably constructed. Even for images that are captured of people responding to commercials or films, those people are aware they are being watched, which can change their responses.

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The difficulty in automating the connection between facial movements and basic emotional categories leads to the larger question of whether emotions can be adequately grouped into a small number of discrete categories at all.⁹⁵

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Based on all the current evidence, the team observed, “It is not possible to confidently infer happiness from a smile, anger from a scowl, or sadness from a frown, as much of current technology tries to do when applying what are mistakenly believed to be the scientific facts.”¹⁰¹

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Once the theory emerged that it is possible to assess internal states by measuring facial movements and the technology was developed to measure them, people willingly adopted the underlying premise. The theory fit what the tools could do. Ekman’s theories seemed ideal for the emerging field of computer vision because they could be automated at scale.

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The more complex issues of context, conditioning, relationality, and cultural factors are hard to reconcile with the current disciplinary approaches of computer science or the ambitions of the commercial tech sector. So Ekman’s basic emotional categories became standard.

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As Barrett writes, “Many of the most influential models in our science assume that emotions are biological categories imposed by nature, so that emotion categories are recognized, rather than constructed, by the human mind.”¹⁰³ AI systems for emotion detection are premised on this idea. Recognition might be the wrong framework entirely when thinking about emotions because recognition assumes that emotional categories are givens, rather than emergent and relational.

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The one-size-fits-all recognition model is not the right metaphor for identifying emotional states. Emotions are complex, and they develop and change in relation to our families, friends, cultures, and histories, all the manifold contexts that live outside of the AI frame. In many cases, emotion detection systems do not do what they claim. Rather than directly measuring people’s interior mental states, they merely statistically optimize correlations of certain physical characteristics among facial images.

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The areas of life in which these systems are operating are expanding as rapidly as labs and corporations can create new markets for them. Yet they all rely on a narrow understanding of emotions—grown from Ekman’s initial set of anger, happiness, surprise, disgust, sadness, and fear—to stand in for the infinite universe of human feeling and expression across space and time. This takes us back to the profound limitations of capturing the complexities of the world in a single classificatory schema.

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The Snowden archive reveals a parallel AI sector, one developed in secrecy. The methods share many similarities, but there are striking differences in terms of the reach, the objectives, and the result. Gone are any rhetorical constructs justifying extraction and capture: every software system is simply described as something to be owned, to be defeated; all data platforms are fair game, and very little is designated as protected.

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From the point of view of the state, the turn to big data and machine learning expanded the modes of information extraction and informed a social theory of how people can be tracked and understood: you shall know them by their metadata. Who is texted, which locations are visited, what is read, when devices spring into action and for what reason—these molecular actions became a vision of threat identification and assessment, guilt or innocence. Harvesting and measuring large aggregates of data at a distance became the preferred way to develop alleged insights into groups and communities as well as assessments of potential targets for killing.

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The AI industry is simultaneously challenging and reshaping the traditional role of states while also being used to shore up and expand older forms of geopolitical power. Algorithmic governance is both part of and exceeds traditional state governance. To paraphrase the theorist Benjamin Bratton, the state is taking on the armature of a machine because the machines have already taken on the roles and register of the state.

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As media scholars such as Wendy Chun and Tung-Hui Hu have noted, the liberal vision of global digital citizens engaging as equals in the abstract space of networks has shifted toward a paranoid vision of defending a national cloud against the racialized enemy.

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Silicon Valley had already built and monetized the logics and infrastructures of AI required to drive a new offset. But first the tech sector had to be convinced that partnering on creating the infrastructure of warfare would be worth it without alienating their employees and deepening public mistrust.

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In April 2017, the Department of Defense published a memo announcing the Algorithmic Warfare Cross-Functional Team, code-named Project Maven.²⁶ “The Department of Defense must integrate artificial intelligence and machine learning more effectively across operations to maintain advantages over increasingly capable adversaries and competitors,” wrote the deputy defense secretary.

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The Defense Department decided to pay tech companies to analyze military data collected from satellites and battlefield drones in places where U.S. domestic privacy laws did not apply. This would align military and U.S. tech sector financial interests around AI without directly triggering constitutional privacy tripwires, as the National Security Agency had done. A bidding war began among the technology companies that wanted the Maven contract, including Amazon, Microsoft, and Google.

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After the fallout from Maven faded, Google’s chief legal officer, Kent Walker, said that the company was pursuing higher security certifications in order to work more closely with the Defense Department. “I want to be clear,” he said. “We are a proud American company.”⁴⁵ Articulating patriotism as policy, tech companies are increasingly expressing strong alignment with the interests of the nation-state, even as their platforms and capacities exceed traditional state governance.

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The underlying worldview built into Palantir’s tools is reminiscent of the NSA: collect everything, then look for anomalies in the data. However, while the NSA’s tools are built to surveil and target enemies of the state, in either conventional or covert warfare, Palantir’s approach has been directed against civilians. As described in a major investigation by Bloomberg in 2018, Palantir is “an intelligence platform designed for the global War on Terror” that is now “weaponized against ordinary Americans at home”.

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As law professor Andrew Ferguson explains, “We are moving to a state where prosecutors and police are going to say ‘the algorithm told me to do it, so I did, I had no idea what I was doing.’ And this will be happening at a widespread level with very little oversight.”

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The shift from traditional to big data surveillance is associated with a migration of law enforcement operations toward intelligence activities. The basic distinction between law enforcement and intelligence is as follows: law enforcement typically becomes involved once a criminal incident has occurred. Legally, the police cannot undertake a search and gather personal information until there is probable cause. Intelligence, by contrast, is fundamentally predictive. Intelligence activities involve

gathering data; identifying suspicious patterns, locations, activity, and individuals; and preemptively intervening based on the intelligence acquired.

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Despite the stated intent of the point system to avoid legally contestable bias in police practices, it hides both intentional and unintentional bias in policing and creates a self-perpetuating cycle: if individuals have a high point value, they are under heightened surveillance and therefore have a greater likelihood of being stopped, further increasing their point value. Such practices hinder the ability of individuals already in the criminal justice system from being further drawn into the surveillance net, while obscuring the role of enforcement in shaping risk scores.

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Given how often governments are turning to contractors to provide the algorithmic architectures for state decision-making, be it policing or welfare systems, there is a case that technology contractors like Palantir should be liable for discrimination and other violations. Currently most states attempt to disclaim any responsibility for problems created by the AI systems they procure, with the argument that “we cannot be responsible for something we don’t understand.” This means that commercial algorithmic systems are contributing to the process of government decision making without meaningful mechanisms of accountability.

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Like a network of private detectives, Vigilant creates a God’s-eye view of America’s interlaced roads and highways, along with everyone who travels along them, while remaining beyond any meaningful form of regulation or accountability.

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As the Snowden documents showed, during the Obama years, the National Security Agency’s global metadata surveillance program would geolocate a SIM card or handset of a suspect, and then the U.S. military would conduct drone strikes to kill the individual in possession of the device.⁷⁹ “We kill people based on metadata,” said General Michael Hayden, former director of the NSA and the CIA.⁸⁰ The NSA’s Geo Cell division was reported to use more colorful language: “We track ’em, you whack ’em.”⁸¹ Signature strikes may sound precise and authorized, implying a true mark of someone’s identity. But in 2014, the legal organization Reprieve published a report showing that drone strikes attempting to kill 41 individuals resulted in the deaths of an estimated 1,147 people.

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These military and policing logics are now suffused with a form of financialization: socially constructed models of creditworthiness have entered into many AI systems, influencing everything from the ability to get a loan to permission to cross borders. Hundreds of such platforms are now in use around the world, from China to Venezuela to the United States, rewarding predetermined forms of social behavior and penalizing those who do not conform.⁸⁵ This “new regime of moralized social classification,” in the words of sociologists Marion Fourcade and Kieran Healy, benefits the “high achievers” of the traditional economy while further disadvantaging the least privileged populations.⁸⁶ Credit scoring, in the broadest sense, has become a place where the military and commercial signatures combine.

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The motifs of scoring and risk have permeated deeply through the structures of state bureaucracy, and the automated decision systems that are imagined in those institutions drive that logic deeply into the way that communities and individuals are imagined, evaluated, scored, and served.

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Phil Agre once observed that “technology at present is covert philosophy; the point is to make it openly philosophical.”⁹⁰ The philosophy here is that data should be extracted globally and structured in order to maintain U.S. hegemony.

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The overlapping grids of planetary computation are complex, cross-breeding corporate and state logics, exceeding traditional state border and governance limits, and they are far messier than the idea of winner takes all might imply. As Benjamin Bratton argues, “The armature of planetary-scale computation has a determining logic that is self-reinforcing if not self-fulfilling, and which through the automation of its own infrastructural operations, exceeds any national designs even if it is also used on their behalf.”⁹¹ The jingoistic idea of sovereign AI, securely contained within national borders, is a myth. AI infrastructure is already a hybrid, and as Hu argues, so is the labor force underpinning it, from factory laborers in China who make electronic components to Russian programmers providing cloud labor to Moroccan freelancers who screen content and label images.

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states are making deals with technology companies they can’t control or even fully understand, and technology companies are taking on state and extrastate functions that they are ill-suited to fulfill and for which, at some point in the future, they might be held liable.

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All the money and resources spent on relentless surveillance is part of a fever dream of centralized control that has come at the cost of other visions of social organization.

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The NSA's distinctive methods and tools have filtered down to classrooms, police stations, workplaces, and unemployment offices. It is the result of enormous investments, of de facto forms of privatization, and the securitization of risk and fear.

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This shift brings with it a different vision of state sovereignty, modulated by corporate algorithmic governance, and it furthers the profound imbalance of power between agents of the state and the people they are meant to serve.

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This epistemological flattening of complexity into clean signal for the purposes of prediction is now a central logic of machine learning. The historian of technology Alex Campolo and I call this enchanted determinism: AI systems are seen as enchanted, beyond the known world, yet deterministic in that they discover patterns that can be applied with predictive certainty to everyday life.⁸ In discussions of deep learning systems, where machine learning techniques are extended by layering abstract representations of data on top of each other, enchanted determinism acquires an almost theological quality.

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The core issue is the deep entanglement of technology, capital, and power, of which AI is the latest manifestation. Rather than being inscrutable and alien, these systems are products of larger social and economic structures with profound material consequences.

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This book proposes that the real stakes of AI are the global interconnected systems of extraction and power, not the technocratic imaginaries of artificiality, abstraction, and automation. To understand AI for what it is, we need to see the structures of power it serves.

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Datasets in AI are never raw materials to feed algorithms: they are inherently political interventions. The entire practice of harvesting data, categorizing and labeling it, and then using it to train systems is a form of politics. It has brought a shift to what are called operational images—representations of the world made solely for machines.¹⁸ Bias is a symptom of a deeper affliction: a far-ranging and centralizing normative logic that is used to determine how the world should be seen and evaluated.

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What epistemological violence is necessary to make the world readable to a machine learning system? AI seeks to systematize the unsystematizable, formalize the social, and convert an infinitely complex and changing universe into a Linnaean order of machine-readable tables. Many of AI's achievements have depended on boiling things down to a terse set of formalisms based on proxies: identifying and naming some features while ignoring or obscuring countless others. To adapt a phrase from philosopher Babette Babich, machine learning exploits what it does know to predict what it does not know: a game of repeated approximations.

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Datasets are also proxies—stand-ins for what they claim to measure. Put simply, this is transmuting difference into computable sameness. This kind of knowledge schema recalls what Friedrich Nietzsche described as “the falsifying of the multifarious and incalculable into the identical, similar, and calculable.”²⁰ AI systems become deterministic when these proxies are taken as ground truth, when fixed labels are applied to a fluid complexity.

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If AI currently serves the existing structures of power, an obvious question might be: Should we not seek to democratize it? Could there not be an AI for the people that is reoriented toward justice and equality rather than industrial extraction and discrimination? This may seem appealing, but as we have seen throughout this book, the infrastructures and forms of power that enable and are enabled by AI skew strongly toward the centralization of control. To suggest that we democratize AI to reduce asymmetries of power is a little like arguing for democratizing weapons manufacturing in the service of peace. As Audre Lorde reminds us, the master's tools will never dismantle the master's house.

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To date, one common industry response has been to sign AI ethics principles. As European Union parliamentarian Marietje Schaake observed, in 2019 there were 128 frameworks for AI ethics in Europe alone.²⁶ These documents are often presented as products of a “wider consensus” on AI ethics. But they are overwhelmingly produced by economically developed countries, with little representation from Africa, South and Central America, or Central Asia. The voices of the people most harmed by AI systems are largely missing from the processes that produce them.

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As Shannon Mattern has noted, the focus is more commonly on the ethical ends for AI, without assessing the ethical means of its application.²⁸ Unlike medicine or law, AI has no formal professional governance structure or norms—no agreed-upon definitions and goals for the field or standard protocols for enforcing ethical practice.

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Tech companies rarely suffer serious financial penalties when their AI systems violate the law and even fewer consequences when their ethical principles are violated. Further, public companies are pressured by shareholders to maximize return on investment over ethical concerns, commonly making ethics secondary to profits. As a result, ethics is necessary but not sufficient to address the fundamental concerns raised in this book.

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To understand what is at stake, we must focus less on ethics and more on power. AI is invariably designed to amplify and reproduce the forms of power it has been deployed to optimize. Countering that requires centering the interests of the communities most affected.³¹ Instead of glorifying company founders, venture capitalists, and technical visionaries, we should begin with the lived experiences of those who are disempowered, discriminated against, and harmed by AI systems.

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The gap between how technology is judged on its release and its lasting consequences has only widened. The social contract, to the extent that there ever was one, has brought a climate crisis, soaring wealth inequality, racial discrimination, and widespread surveillance and labor exploitation. But the idea that these transformations occurred in ignorance of their possible results is part of the problem. The philosopher Achille Mbembé sharply critiques the idea that we could not have foreseen

what would become of the knowledge systems of the twenty-first century, as they were always “operations of abstraction that claim to rationalize the world on the basis of corporate logic.”³³ He writes: “It is about extraction, capture, the cult of data, the commodification of human capacity for thought and the dismissal of critical reason in favour of programming. . . . Now more than ever before, what we need is a new critique of technology, of the experience of technical life.”

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Refusal requires rejecting the idea that the same tools that serve capital, militaries, and police are also fit to transform schools, hospitals, cities, and ecologies, as though they were value neutral calculators that can be applied everywhere.

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Above all, I see the greatest hope in the growing justice movements that address the interrelatedness of capitalism, computation, and control: bringing together issues of climate justice, labor rights, racial justice, data protection, and the overreach of police and military power.